5.2 Logarithmic Functions and Their Graphs

Name _____ Period ____ Date _____

Pre-Calculus 5.2 (30-minute plan) · Student sheet

I can…

- rewrite between logarithmic and exponential form.
- evaluate logarithms and use their basic properties.
- graph logarithmic functions and find their domains.

Vocabulary · fill in the missing word

Logarithmic function — $y =$ _____ $= a^y$ $(a > 0, a \neq 1)$

Common logarithm — $\log x =$ _____

Natural logarithm — $\ln x =$ _____

Part 1 · Notes and examples (fill in each note, then work the example)

Definition: $y = \log_a x$ if and only if _____.

Example 1

a. Write $\log_2 8 = 3$ and $\log_5(\frac{1}{25}) = -2$ in exponential form.

b. Write $4^3 = 64$ in logarithmic form.

Example 2

Evaluate without a calculator.

a. $\log_2 32$ b. $\log_3 1$ c. $\log_4 2$ d. $\log(\frac{1}{100})$

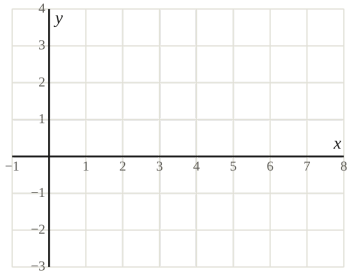
Properties: $\log_a 1 = 0$, $\log_a a = 1$, $\log_a a^x = x$, and a^($\log_a x$) = _____.

Graph of $y = \log_a x$: x-intercept (1, 0), vertical asymptote _____, domain $x > 0$.

Domain: the expression inside a logarithm must be _____.

Example 3

Graph $f(x) = \log_3 x$ and $g(x) = \log_3(x - 2) + 1$. Give each asymptote and x-intercept.



Part 2 · Partner practice

Partner practice

Find the domain and vertical asymptote of $f(x) = \log_4(x + 5)$.